# High Dimensional Expression Matrix

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## ABSTRACT

Understanding complex relationships in high-dimensional data is a significant challenge. While many visualization techniques focus on global structure or projections, the nuanced ways data points relate to each other across varying combinations of features (subspaces) often remain obscured. We introduce HEIDI, a novel visualization system that systematically analyzes primarily k-nearest neighbor (k-NN) relationships across the entire powerset of feature subspaces. HEIDI constructs an n x n matrix where each cell (i, j) contains a minimal boolean expression summarizing the specific subspaces in which point j is a k-NN of point i. This "expression matrix" provides an interpretable, qualitative descriptor of pairwise interactions. We present visualization techniques for this matrix, including cluster-aware sorting for highlighting intra- and inter-cluster patterns, and complementary visualizations that link expressions to data geometry and network structures. HEIDI offers a unique perspective by not just identifying if points are neighbors, but under what feature conditions they are neighbors, thus providing deeper insights into the underlying data structure. We demonstrate its utility on synthetic and real-world datasets, highlighting its ability to reveal complex, subspace-specific relationships, and discuss future work on hierarchical analysis and enhanced interactivity.

## I. INTRODUCTION

The proliferation of high-dimensional datasets across diverse domains, from bioinformatics to finance, necessitates effective tools for exploration and understanding. A critical aspect of this understanding lies in how data points relate to each other, particularly how these relationships change when considering different subsets of features or dimensions. Traditional methods often rely on global distance metrics or dimensionality reduction techniques, which can obscure localized, subspace-specific patterns. For instance, two data points might be distant in the full feature space but very close in a specific 2D or 3D subspace that defines a particular context or function.

While subspace clustering and analysis methods [1, 2] have aimed to identify clusters within various subspaces, a significant challenge remains in visualizing and interpreting the precise nature of these subspace-specific relationships for all pairs of points, especially how these diverse interactions contribute to the formation and differentiation of clusters. Existing visualization tools might show projections onto selected subspaces, but a systematic, interpretable overview of pairwise relationships across a multitude of subspaces, particularly one that explicitly encodes the conditions of these relationships, is lacking.

This paper introduces HEIDI, a visualization system designed to address this gap by centering on a novel Interpretable Expression Matrix. HEIDI's core contributions are:

**Systematic Subspace k-NN Analysis:** It comprehensively computes k-nearest neighbor (k-NN) relationships for every pair of points across every non-empty subset of features (the powerset of dimensions).

**The Interpretable Expression Matrix:** This is HEIDI's central innovation. It generates an n x n matrix where each cell (i, j) is populated not with a scalar similarity value, but with a minimal boolean expression. This expression (e.g., "F0 + F3") concisely denotes the specific feature subspaces (e.g., features F0 or F3) in which point j is among the k-nearest neighbors of point i. This matrix provides an unprecedented, rich tapestry of pairwise interactions, explicitly qualified by their subspace origins.

**Multi-faceted Visualization Suite Focused on Matrix Interpretation:** HEIDI provides powerful visualizations of this expression matrix, notably a sorted heatmap that, when aligned with known cluster structures, vividly reveals dominant subspace expressions characterizing both intra-cluster coherence and inter-cluster interactions. Complementary tools connect these abstract expressions back to the data's geometric layout (e.g., scatter plots with expression-colored links) and network structures, all anchored to the insights derived from the expression matrix.

HEIDI fundamentally shifts the analytical focus from merely whether points are neighbors to why and under what precise feature conditions they are neighbors. The Expression Matrix, as its primary artifact, offers a richer, more qualitative, and highly granular understanding of high-dimensional data structure, making explicit the diverse subspace-driven forces that shape both the internal cohesion of clusters and the distinct relationships between them.

## II. METHODS

The HEIDI system comprises several stages: data preprocessing, HEIDI matrix computation, and visualization.

**Data Preprocessing and k-NN Computation**

Given a dataset X with n samples and d features, data is typically normalized. The core computation involves, for each feature subspace s in the powerset P(D) of dimensions D, and for each point X_i, finding its k nearest neighbors within that subspace s. This results in a set of k-NN lists, one for each point-subspace pair.

**HEIDI Matrix Construction and Expression Generation**

From the k-NN computations, we construct an intermediate n x n x |P(D)| binary tensor H_binary, where H_binary[i, j, s_idx] = 1 if X_j is a k-NN of X_i in subspace s (indexed by s_idx), and 0 otherwise.

For each pair of points (X_i, X_j), the vector v_{ij} = H_binary[i, j, :] captures their k-NN relationship across all subspaces. The HEIDI system then converts this binary vector v_{ij} into a minimal boolean expression. For example, if v_{ij} has 1s at indices corresponding to feature F0 and feature F3 only, the expression is "F0 + F3". If v_{ij} is all zeros, the expression is null (no k-NN relationship in any subspace). This results in the n x n HEIDI Expression Matrix, where each cell (i,j) contains such an expression.

**Visualization Techniques**

Given the HEIDI Expression Matrix, we provide several coordinated visualizations:

**Sorted HEIDI Matrix Heatmap (plot_improved_heidi_matrix):**

Data points are first sorted by their cluster labels y to group similar points.

The n x n matrix is visualized as a heatmap, where each cell (i,j) is colored uniquely based on its minimal expression. This reveals dominant expressions within intra-cluster blocks (on the diagonal) and inter-cluster blocks (off-diagonal).

An **improved_anti_diagonal_sort** algorithm is applied within each cluster block to further organize expressions, often highlighting dominant patterns (e.g., anti-diagonals if points in a sequence share similar expressions with their predecessors/successors).

A legend displays the most frequent expressions and their corresponding colors.

**Expression Connection Scatter Plots (visualize_expression_connections_enhanced):** Data points are plotted in a 2D projection (e.g., using the first two features, or MDS/PCA of the full data).

Lines are drawn between pairs of points (X_i, X_j) if their corresponding HEIDI cell (i,j) contains one of the top N most frequent expressions. Lines are colored by the expression.

This visualization links the abstract expressions to the geometric distribution of points, showing which types of subspace relationships connect which points or clusters. Individual plots per expression and summary plots showing average cluster-to-cluster connections are generated.

**Expression Network Visualization (visualize_expression_network):**

Points are represented as nodes, colored by cluster.

Edges are drawn between points (X_i, X_j) if they share a top N HEIDI expression, with edges colored by the expression.

Standard graph layout algorithms (e.g., spring, circular) are used to reveal the connectivity structure induced by these subspace relationships.

**Sector-Based Expression Analysis (visualize_expressions_by_sector):**

The 2D data projection is divided into spatial sectors (e.g., along one axis).

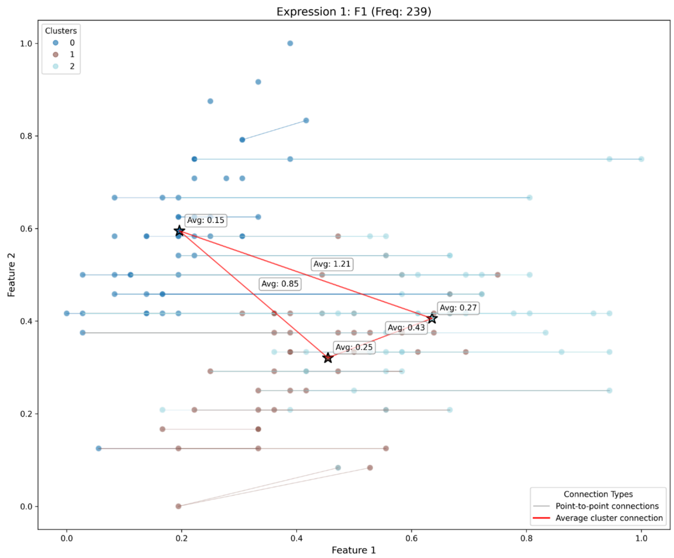
For each sector, a table details the points within it and which of the top N HEIDI expressions are associated with connections involving these points. This helps understand spatial variations in expression patterns.

## III. RESULTS AND DISCUSSION

### Iris Dataset:

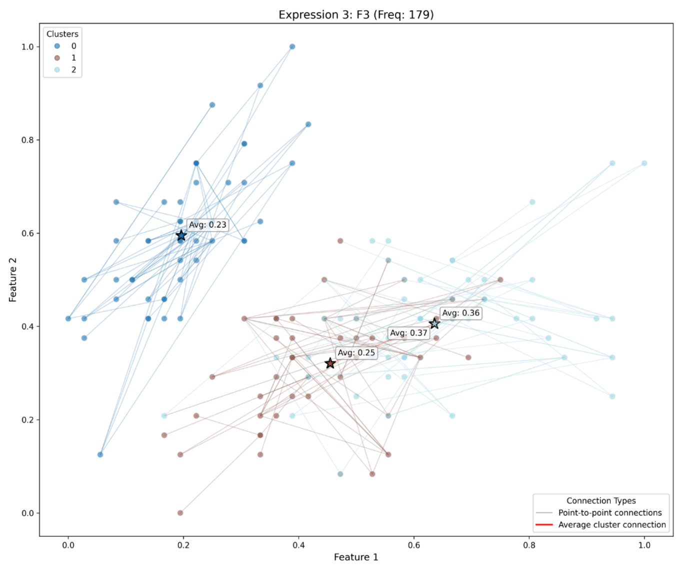
Analysis of the Iris dataset reveals characteristic expressions for each species, highlighting the specific feature combinations (e.g., petal length + sepal width) that best define nearest-neighbor relationships within each flower type and distinguish between them. The connection plots show how these expressions link points within and sometimes across species boundaries.

Figure 1 Shows the actual Matrix along with the improved Anti Diagonal Pattern in which we can



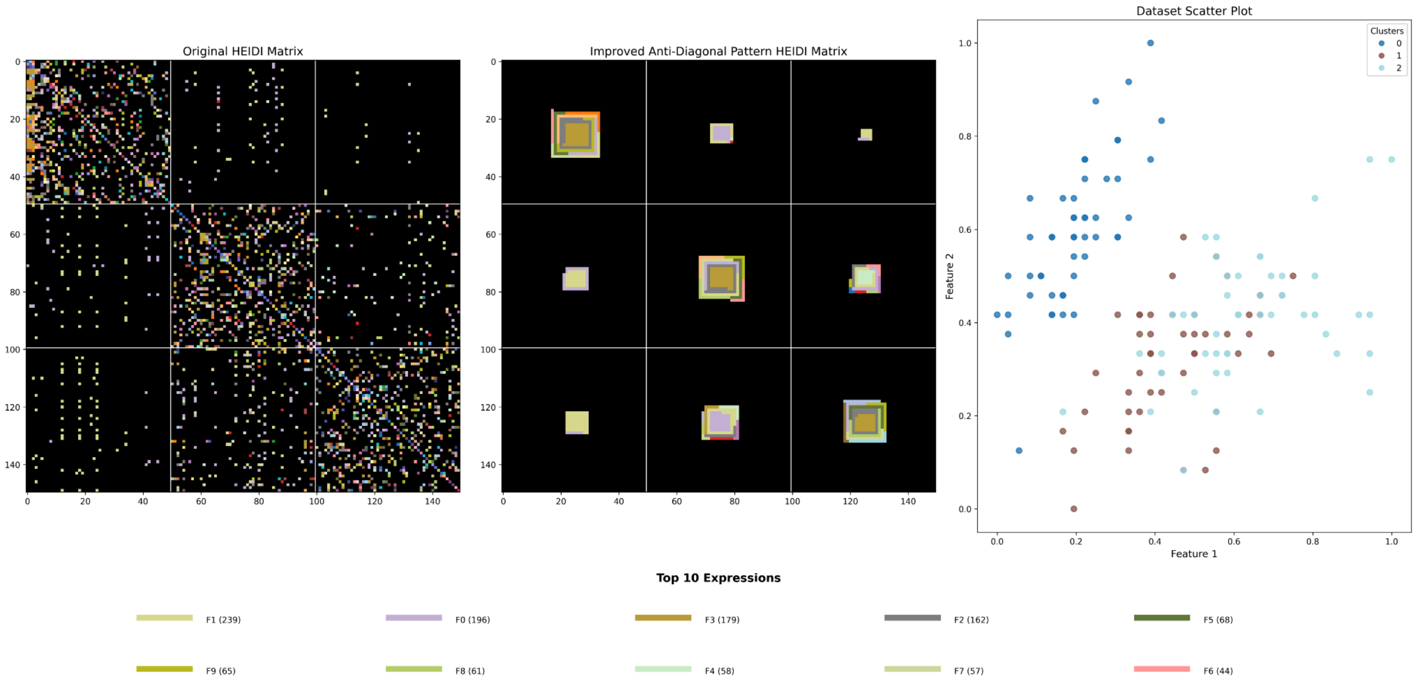

see the majority of expressions and how do inter cluster and intra cluster expressions fare within diagonal and non diagonal blocks respectively.

We can see F3 being majority expression in diagonal hence intra cluster pairs of points and that shows in the linkage network in Figure 3. Similarly F1 being majority in non diagonal hence



inter cluster pairs of points is shown in a similar vein in Figure 2 that has majority connections that go beyond the clusters.

Figure 1 overall shows which expressions lie in each of the blocks that constitute the inter and intra cluster relationships.

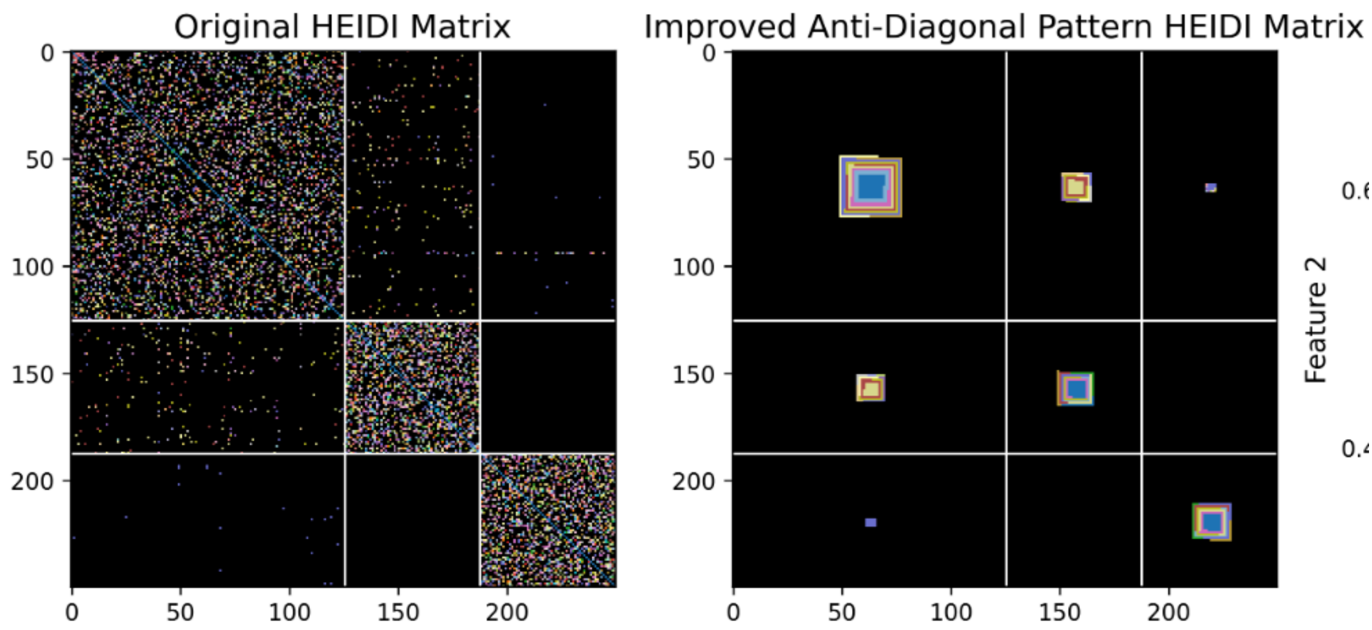


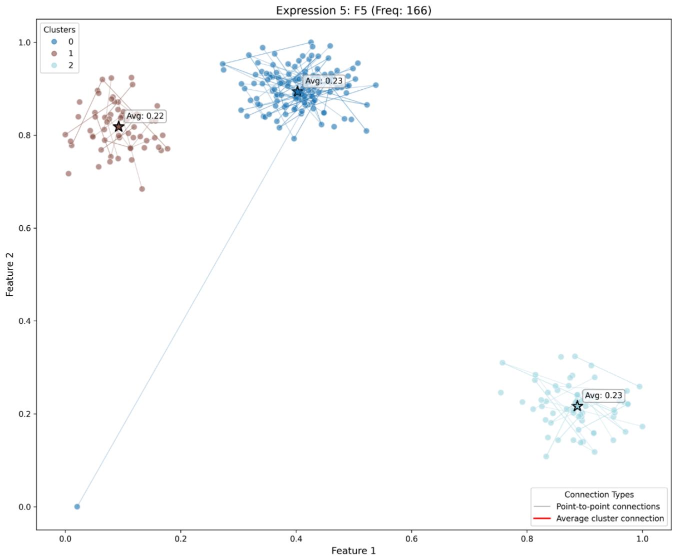

**Synthetic Data:** On synthetic datasets with known subspace clusters, HEIDI successfully identifies the correct subspace expressions defining intra-cluster coherence and inter-cluster separation, demonstrating its ability to recover ground truth.

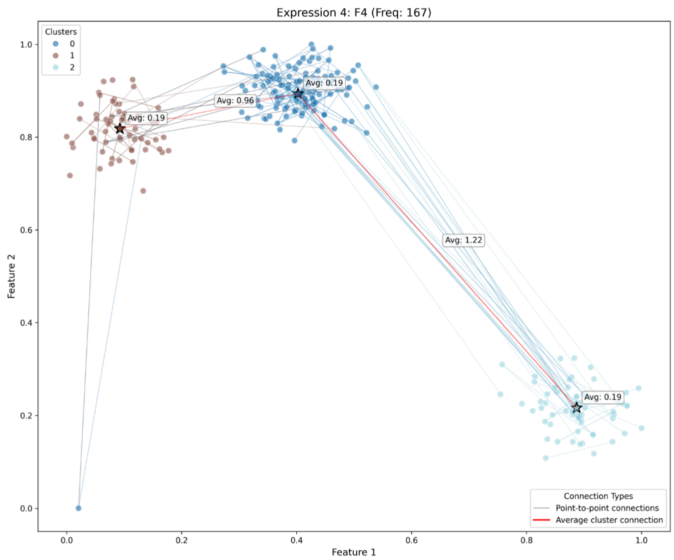
The Matrix in Figure 4 shows the Diagonal Blocks having F5 as the majority of Expressions as can be seen in Figure 5.

The Purple block in block 1,3 and 3,1 show that expression F4 has a lot of Nearest Neighbors link between cluster 1 and 3 and is validated by the linkage graphs.

These examples show how intuitively HEIDI can help visualise relationships between different subspaces amongst different intra and inter cluster pairs of points.






### How HEIDI is Better: Advantages and Insights

**Interpretability of Relationships:** Instead of just knowing points are "close," HEIDI tells us why (in which specific feature subspaces). Expressions like "F0+F1" (neighbors in the F0-F1 subspace) are directly interpretable.

**Comprehensive Subspace Analysis:** By considering the powerset of dimensions, HEIDI provides a more exhaustive view of subspace relationships than methods relying on selected projections or limited subspace searches.

**Qualitative Understanding:** The focus on symbolic expressions provides a qualitative understanding of pairwise interactions, complementing quantitative distance measures.

**Revealing Hidden Structures:** HEIDI can reveal relationships that are not apparent in the full feature space or in any single low-dimensional projection. For example, two distinct clusters might appear overlapped in full space but be clearly separated and internally coherent within different, specific subspaces, which HEIDI's expressions would highlight.

**Cluster Characterization:** The dominant expressions within and between clusters can characterize the nature of those clusters and their interactions (e.g., "Cluster A points are strongly self-similar in F1,F2, while Cluster B points are self-similar in F3,F4. They interact primarily via subspace F0.").

## IV. LIMITS AND FUTURE WORK

### Current Limitations

**Scalability with Number of Features (Dimensionality):** The core strength of HEIDI—analyzing the powerset of dimensions—is also its primary bottleneck. The 2^d growth of the subspace search space leads to exponential computational and memory demands as the number of features (d) increases. This practically limits the number of features that can be exhaustively analyzed.

**Scalability with Number of Samples (n):** The n x n HEIDI Expression Matrix becomes very large for datasets with many samples. This impacts memory, computation for sorting, and visualization performance. Furthermore, with very large n, individual pairwise expressions might become sparse, potentially making global patterns harder to discern without aggregation.

**Interpretability with High Label Cardinality:** When datasets have a large number of distinct labels (clusters), the sorted HEIDI matrix can fragment into an excessive number of small inter- and intra-cluster blocks. This can dilute the visual impact of dominant expressions and reduce the clarity of connections between specific cluster pairs.

**Complexity of Connection Visualizations:**

**Projection Dependency:** Current network connection plots (visualize_expression_connections_enhanced, visualize_expression_network) are often overlaid on 2D projections of the data. The utility of these connections is thus dependent on the quality of the 2D projection, which may not always effectively represent the true high-dimensional relationships relevant to the expressions.

**Visual Clutter:** For dense datasets or highly prevalent expressions, the number of connections drawn can be overwhelming, leading to significant visual clutter and obscuring underlying patterns.

**Proposed Improvements and Future Directions**

To address current limitations and enhance HEIDI's capabilities, we propose several interconnected areas for future work:

**Scalability Enhancements:**

**Feature Scalability:** Mitigate the 2^d subspace explosion by implementing intelligent subspace pruning (e.g., limiting subspace cardinality [3], feature ranking/prioritization) and enabling interactive or progressive subspace exploration. Concepts from sparse TDA could also guide the identification of significant low-dimensional embeddings.

**Sample Scalability:** Address large n by developing matrix aggregation and summarization techniques (e.g., aggregating expressions for point groups, focusing on meta-expressions, point/pair sampling). Explore alternative, more compact matrix representations like bipartite graphs linking point-groups to expressions.

**Improving Interpretability and Visual Clarity:**

**High Label Cardinality:** Introduce hierarchical cluster views in the HEIDI matrix, allowing interactive aggregation of cluster blocks for a multi-level overview, complemented by metrics to highlight key inter-cluster expressions.

**Connection Visualization Overhaul – Hierarchical Linkage Analysis:** To combat visual clutter and projection dependency in connection plots, we propose a Hierarchical Expression Linkage Visualization. Inspired by topological simplification and Pareto optimality concepts [4], this will:

- Define multi-objective criteria for "dominant connections" (based on connectivity degree, geometric properties, point centrality, expression purity).
- Hierarchically display only top-tier "Pareto-optimal" or "Skyline" connections, with interactive drill-down to reveal further layers, potentially using projection-agnostic abstract graph layouts.
- Parallel Coordinates can be used to reduce the similar points for better and cleaner linkages.

**Refining Expression Significance and Interpretation:**

**Beyond Frequency:** Develop metrics for expression "significance" beyond simple occurrence frequency (e.g., cluster-specificity, geometric coherence in relevant subspaces, statistical significance).

**Hierarchical Expression Summarization:** Implement methods to group similar expressions into "meta-expressions," simplifying legend and matrix interpretation.

**Leveraging Topological Data Analysis (TDA) Concepts:** Explore the application of TDA principles [4] to further abstract and summarize HEIDI's output. For instance, build simplicial complexes where simplices represent co-occurring features in expressions or groups of points sharing similar expression profiles. Analyzing the persistent homology of such complexes could reveal robust, multi-scale structural patterns in the subspace relationships that HEIDI uncovers, offering a more theoretically grounded approach to simplification and feature definition.

**Enhanced Interaction and Exploration (Inspired by Yuan et al. [3]):**
Implement dynamic filtering and linking across all views, triggered by user selections of expressions or matrix cells.
Provide on-demand generation of standard 2D/3D projections based on user-selected features or HEIDI expressions for direct visual grounding.
Allow user-guided definition of feature subsets for focused HEIDI re-analysis.

**Strengthening Analytical Depth:**
**Quantitative Metrics:** Integrate quantitative measures associated with expressions (e.g., average k-NN distance within involved subspaces, "separation power" of inter-cluster expressions).
**Integration with Automated Methods:** Leverage HEIDI's outputs (dominant subspaces, significant expressions) as priors or features for downstream machine learning tasks like classification, semi-supervised learning, or targeted subspace clustering, guiding the parameterization of these automated methods.

## V. CONCLUSION

This paper introduced HEIDI (High-dimensional Exploration via Ideated Descriptors), a novel visualization system centered on an Interpretable Expression Matrix designed to unravel complex, subspace-specific relationships in high-dimensional data. By systematically analyzing k-nearest neighbor connections across the powerset of feature subspaces and representing these as minimal boolean expressions, HEIDI shifts the focus from mere proximity to the explicit feature conditions defining these pairwise interactions. Our multi-faceted visualization suite, particularly the sorted heatmap of the expression matrix, alongside linked scatter plots and network views, effectively reveals dominant subspace expressions that characterize both intra-cluster coherence and inter-cluster dynamics. Through case studies on synthetic and real-world datasets like the Iris dataset, we demonstrated HEIDI's ability to provide interpretable, qualitative insights into how data points relate within specific feature contexts, often uncovering structures not apparent in global views. The Expression Matrix serves as a powerful artifact for understanding the diverse subspace-driven forces shaping data organization.

While HEIDI offers significant advantages in interpretability and comprehensive subspace analysis, we acknowledge limitations related to scalability with feature and sample sizes, as well as the visual complexity associated with high label cardinality and dense connection plots. Future work will focus on addressing these challenges through intelligent subspace pruning, matrix aggregation techniques, and, crucially, the development of a Hierarchical Expression Linkage Visualization inspired by Pareto optimality and topological simplification. Further enhancements will include refining metrics for expression significance beyond frequency, leveraging TDA concepts for deeper structural abstraction, and fostering richer user interaction for dynamic exploration, as inspired by related works.

Ultimately, HEIDI provides a unique and valuable perspective for high-dimensional data analysis. By continuing to develop its scalability, interpretative power, and interactive capabilities, we believe HEIDI can evolve into an even more potent tool for researchers and analysts seeking to understand the nuanced, multi-faceted relationships hidden within their complex datasets.

## VI. REFERENCES

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